

Exploratory Data Analysis Report

Customer Behaviour Data Visualization Using PCA and t-SNE

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Dataset	ecommerce_customer_behavior.csv
Records	1,500 customers
Raw columns	11 (7 numeric, 2 categorical, ID, true_segment)
Methods	Median imputation, StandardScaler, one-hot encoding, PCA, t-SNE, KMeans (k=4)

1. Executive Summary

- The dataset holds **1,500 customers** and **7 numeric behavioural features** plus two categorical preferences (device and product category). Only 45 values (15 each in three columns) were missing and were filled with the column median.
- The features are **strongly inter-correlated**: purchase frequency vs cart abandonment is $r = -0.82$, and cart abandonment vs days since last purchase is $r = 0.81$. This redundancy is what justifies PCA.
- PCA needs **8 of 13 components** to retain at least 90% of the variance (cumulative 90.7%), matching the project claim. The first two components explain only 40.0%, so a 2D PCA plot is a partial view.
- KMeans with $k=4$ on the PCA space yields segments of 451 (Frequent Buyers), 249 (High-Value), 300 (Discount-Driven) and 500 (Occasional Browsers) customers with clearly different profiles. Silhouette score is **0.239** (moderate-to-weak cohesion).
- The recovered segments agree with the dataset's own `true_segment` column on **99.5%** of customers (adjusted Rand index 0.989).
- **Important correction**: the hard-coded `label_map` in the notebook gives wrong names to three of the four clusters in the exported file (details in Section 7).

2. Dataset Overview

Each row is one customer. The table lists every column, its meaning and type.

Column	Type	Description
<code>customer_id</code>	ID	Unique customer identifier (CUST00001-CUST01500)
<code>purchase_frequency</code>	numeric	Purchases per period
<code>avg_order_value</code>	numeric	Average order value
<code>browsing_time_minutes</code>	numeric	Average browsing time
<code>session_count</code>	numeric (int)	Number of sessions
<code>discount_usage_rate</code>	numeric 0-1	Share of orders using a discount
<code>cart_abandonment_rate</code>	numeric 0-1	Share of carts abandoned
<code>days_since_last_purchase</code>	numeric	Recency in days
<code>preferred_device</code>	categorical	mobile / desktop / tablet
<code>preferred_category</code>	categorical	fashion / electronics / grocery / beauty / home
<code>true_segment</code>	categorical	Reference segment label shipped with the data (not used in clustering)

2.1 Data quality

- **Missing values**: `avg_order_value` (15), `browsing_time_minutes` (15), `days_since_last_purchase` (15) - 1.0% of rows each. Imputed with medians of 35.23, 28.20 and 10.00 respectively. Median is a sensible choice because these three columns are right-skewed.
- **Duplicates**: none. All 1,500 customer IDs are unique.
- **Other columns**: no missing values; categorical columns have no inconsistent spellings.

3. Univariate Analysis

3.1 Summary statistics (raw data, before imputation)

Feature	Count	Mean	Std	Min	25%	50%	75%	Max	Skew
purchase_frequency	1500	9.32	6.76	0.0	2.6	8.6	15.1	28.8	0.43
avg_order_value	1485	47.64	36.56	7.24	23.67	35.23	50.9	181.46	1.7
browsing_time_minutes	1485	29.7	11.39	2.9	21.3	28.2	36.7	67.1	0.54
session_count	1500	15.28	4.88	2.0	12.0	15.0	19.0	33.0	0.41
discount_usage_rate	1500	0.29	0.26	0.0	0.12	0.19	0.36	0.95	1.22
cart_abandonment_rate	1500	0.37	0.26	0.05	0.13	0.34	0.6	0.85	0.34
days_since_last_purchase	1485	19.7	20.53	0.0	4.0	10.0	35.5	88.3	1.08

3.2 Distributions

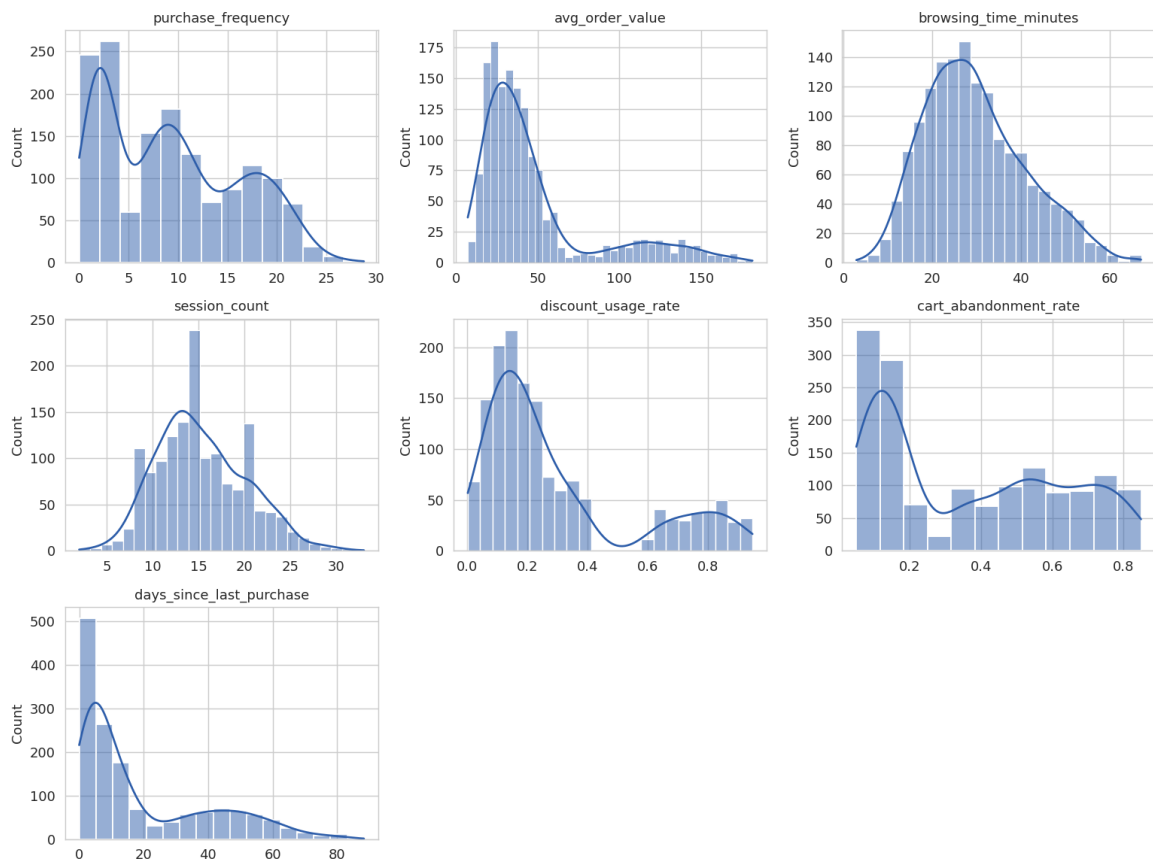


Figure 1. Histograms with KDE for the seven numeric features (after imputation).

- **avg_order_value**, **discount_usage_rate** and **days_since_last_purchase** are right-skewed with long tails.
- **purchase_frequency**, **cart_abandonment_rate** and **browsing_time** show multi-modal or wide shapes, a first hint that the population is a mixture of distinct groups rather than one homogeneous crowd.
- **session_count** is the most symmetric feature (roughly bell-shaped around 15).

3.3 Outliers (IQR rule)

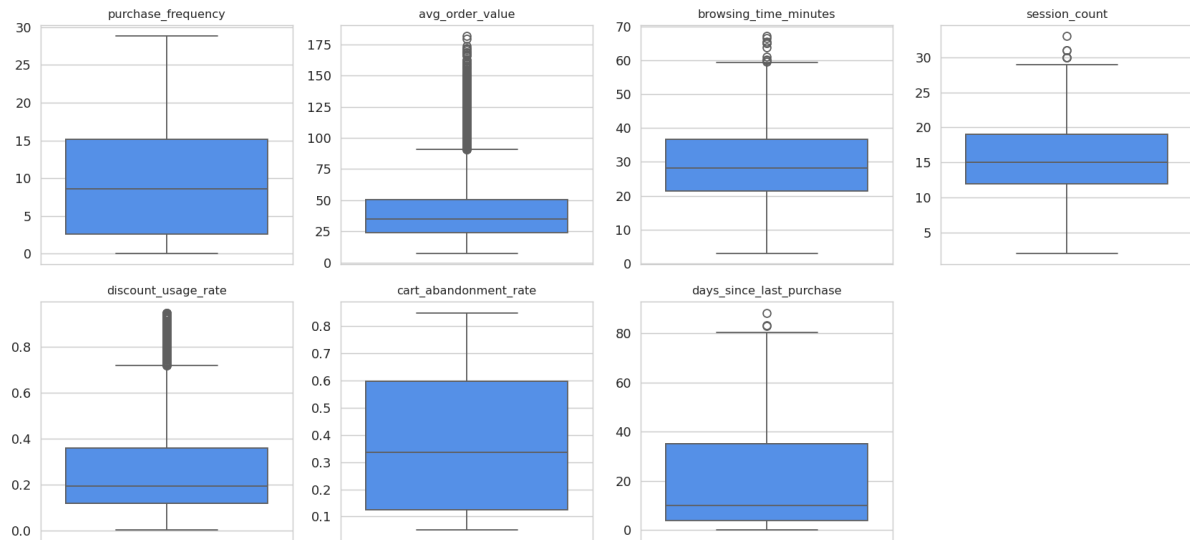


Figure 2. Box plots of the numeric features.

Feature	Points beyond 1.5 x IQR
purchase_frequency	0
avg_order_value	215
browsing_time_minutes	9
session_count	6
discount_usage_rate	202
cart_abandonment_rate	0
days_since_last_purchase	3

- **avg_order_value** (215 points) and **discount_usage_rate** (202 points) hold nearly all the flagged values. These are not errors: they are the High-Value and Discount-Driven customers sitting in the tails, so they were **kept**, not removed.
- The remaining features have 9 or fewer flagged points and need no treatment.

3.4 Categorical features

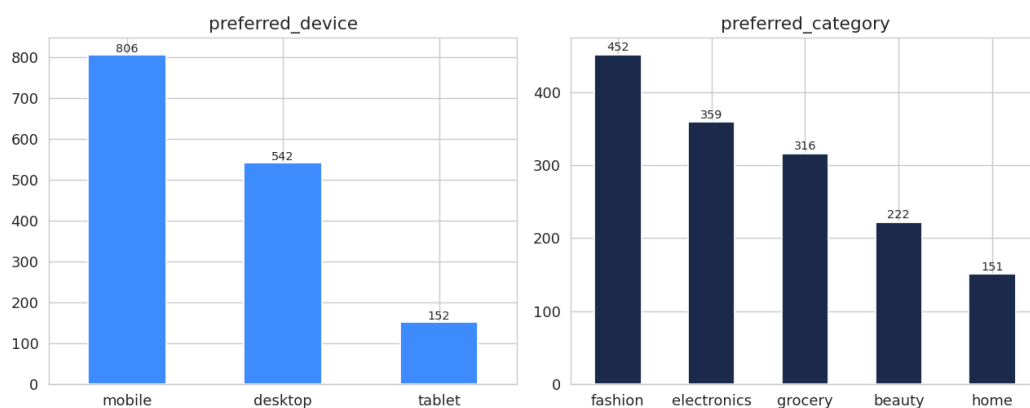


Figure 3. Preferred device and preferred category.

- **Device:** mobile 806 (53.7%), desktop 542 (36.1%), tablet 152 (10.1%).
- **Category:** fashion 452 (30.1%), electronics 359, grocery 316, beauty 222, home 151 (10.1%).

4. Bivariate Analysis

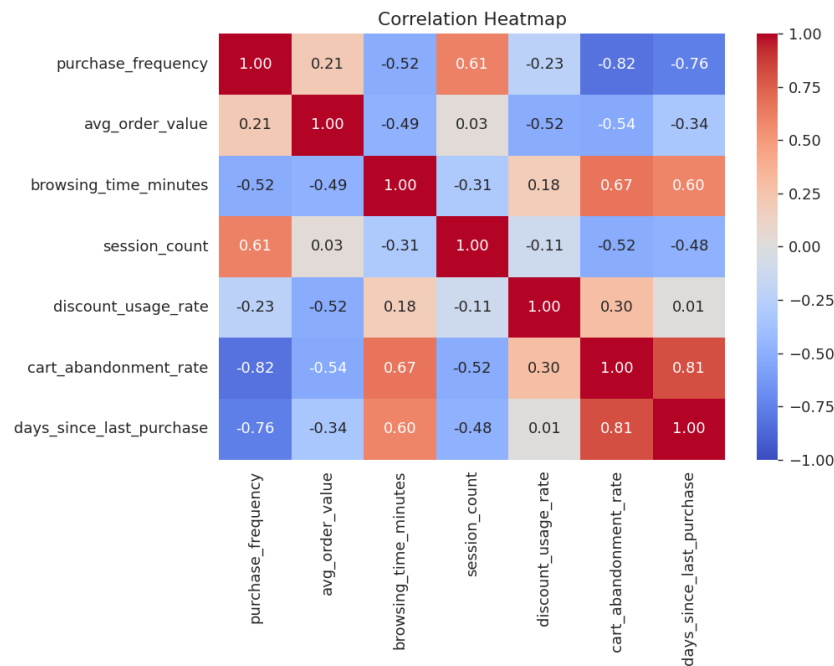


Figure 4. Pearson correlation of the numeric features.

Feature A	Feature B	r
purchase_frequency	cart_abandonment_rate	-0.82
cart_abandonment_rate	days_since_last_purchase	0.81
purchase_frequency	days_since_last_purchase	-0.76
browsing_time_minutes	cart_abandonment_rate	0.67
purchase_frequency	session_count	0.61
browsing_time_minutes	days_since_last_purchase	0.6

- Purchase frequency, cart abandonment and recency form one tight block: customers who buy often abandon few carts and bought recently. Browsing time moves with abandonment and recency.
- Such strong correlations mean many features carry overlapping information, so compressing them with PCA before clustering is well-motivated.

5. Dimensionality Reduction

5.1 Preprocessing

The two categorical columns were one-hot encoded (drop-first), giving 13 features in total (7 numeric + 2 device dummies + 4 category dummies). All features were standardised with StandardScaler before PCA.

5.2 PCA

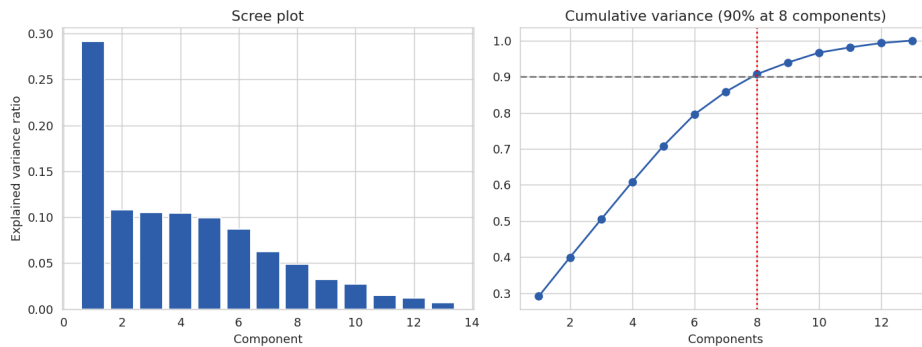


Figure 5. Scree plot and cumulative explained variance.

Component	PC1	PC2	PC3	PC4	PC5	PC6	PC7	PC8	PC9
Explained (%)	29.2	10.8	10.5	10.4	9.9	8.7	6.3	4.9	3.2
Cumulative (%)	29.2	40.0	50.4	60.9	70.8	79.5	85.8	90.7	93.9

- **8 components** are needed to reach 90% (cumulative 90.7%); 7 components give 85.8%.
- PC1 explains 29.2% and PC2 10.8%; after PC1 the variance is spread fairly evenly, so there is no sharp elbow.
- Part of the spread beyond PC2 comes from the one-hot device/category columns, which carry little segment signal (Section 6.3).

5.3 PCA (2D) and t-SNE

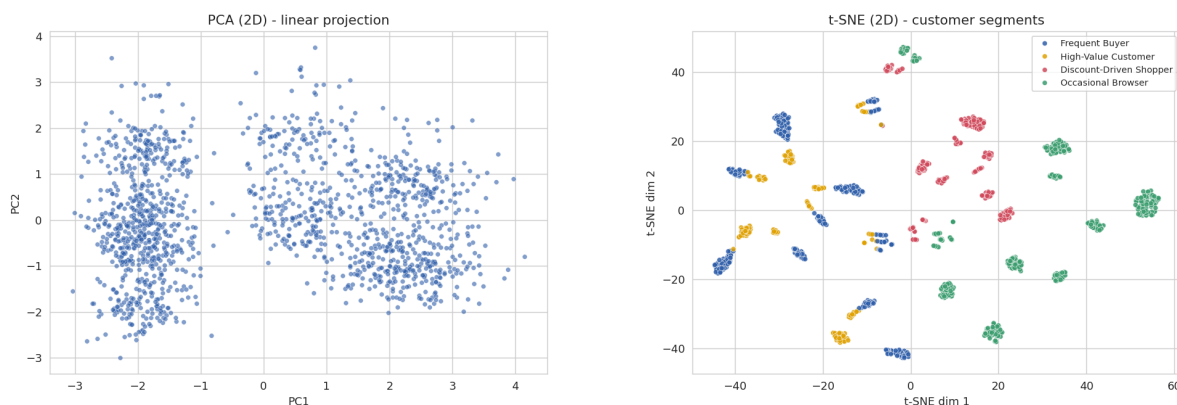
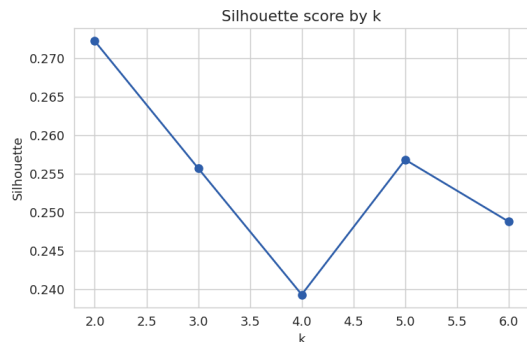


Figure 6. Left: PCA 2D projection (uncoloured). Right: t-SNE (perplexity 35) on the 8-component PCA output, coloured by segment.

- The 2D PCA plot shows a connected cloud with some structure, but it captures only 40% of the variance, so groups overlap.
- t-SNE separates the customers into **four distinct islands**, confirming that non-linear local structure is much clearer than the linear view (island distances and sizes are not directly interpretable).

6. Clustering and Segment Profiles

6.1 Choosing k



k	Silhouette
2	0.272
3	0.256
4	0.239
5	0.257
6	0.249

- Silhouette scores are low for every k (0.24-0.27). k=2 scores highest (0.272), and k=4 (0.239) is the lowest of the five values tested.
- k=4 was kept because it matches the four business segments targeted by the project and the four islands seen in t-SNE; the silhouette score alone would not select it. This should be stated openly in the review.
- The modest silhouette is partly due to noisy dimensions (one-hot columns and weaker PCs) diluting the geometry used by KMeans.

6.2 Segment profiles

Segment	Purch. freq.	Avg order value	Browsing time (min)	Sessions	Discount rate	Cart abandon.	Days since last purchase	Customers	Share (%)
Frequent Buyer	17.95	45.29	24.63	19.99	0.15	0.12	3.16	451	30.1
High-Value Customer	10.0	120.64	17.82	13.98	0.08	0.12	7.04	249	16.6
Discount-Driven Shopper	7.98	20.29	29.88	14.85	0.77	0.42	12.2	300	20.0
Occasional Browser	2.01	29.43	40.04	11.94	0.25	0.68	45.12	500	33.3

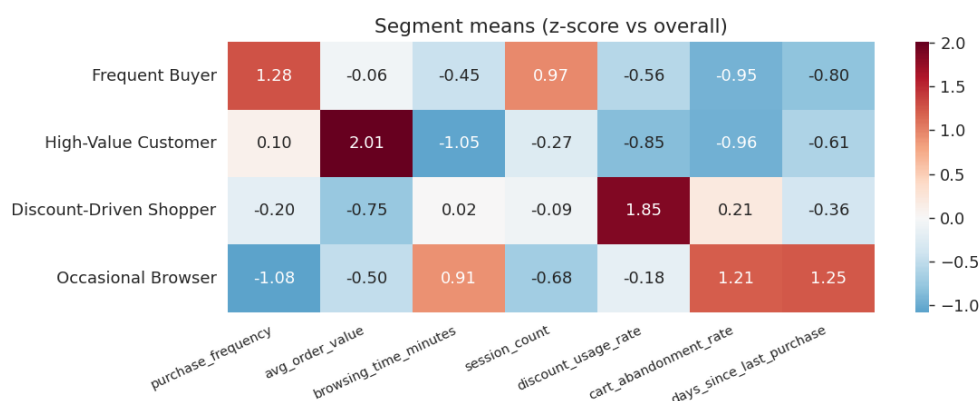


Figure 7. Segment means as z-scores relative to the whole population (red = above average, blue = below).

Segment interpretation

- **Frequent Buyer (451)**: highest purchase frequency (17.95) and sessions (20), lowest cart abandonment (0.12) and most recent purchases (3 days). Average basket size. Action: loyalty rewards and cross-sell.
- **High-Value Customer (249)**: by far the largest orders (about 120.64 vs population mean 47.6), very low discount use (0.08), low abandonment, short browsing time. Action: premium offers and early access, no discounting.
- **Discount-Driven Shopper (300)**: highest discount use (0.77) and smallest basket (about 20.29), moderate frequency. Action: targeted promotions and bundle offers to lift basket size.
- **Occasional Browser (500)**: lowest purchase frequency (2.0), longest browsing (40 min), highest abandonment (0.68), longest gap since last purchase (45 days). Action: re-engagement campaigns and cart-recovery reminders.

6.3 Segment mix by device and category

Segment (% of segment)	desktop	mobile	tablet
Frequent Buyer	36.4	53.2	10.4
High-Value Customer	32.1	60.2	7.6
Discount-Driven Shopper	32.0	57.0	11.0
Occasional Browser	40.4	49.0	10.6

Segment (% of segment)	beauty	electronics	fashion	grocery	home
Frequent Buyer	17.5	23.9	29.9	20.0	8.6
High-Value Customer	15.7	23.7	26.1	24.9	9.6
Discount-Driven Shopper	11.0	23.7	31.0	23.7	10.7
Occasional Browser	14.2	24.2	31.8	18.6	11.2

- Device and category mix is close to the overall distribution in every segment (mobile 49-60%, fashion 26-32%). Segments are therefore defined by **behaviour**, not by device or category.
- Small differences (e.g. High-Value customers are the most mobile-heavy at 60%; Discount-Driven shoppers over-index least on beauty) are worth a follow-up test but are not strong signals.

6.4 Agreement with the dataset's reference segments

The file contains a `true_segment` column that was not used for clustering. After assigning names from the cluster profiles, **99.5%** of customers (about 7 of 1,500 differ) fall in the matching reference segment, adjusted Rand index = 0.989. This shows the pipeline recovers the intended structure almost perfectly. The dataset appears to be synthetic or simulated, so this result should not be read as a guarantee on real-world data (the project's own future scope already notes validating on a public real dataset).

7. Issue Found: Cluster Label Mapping

The notebook assigns names with a fixed dictionary (0 = Frequent Buyer, 1 = High-Value Customer, 2 = Discount-Driven Shopper, 3 = Occasional Browser). The notebook's own comment warns that IDs can change between runs, and in the exported file the mapping does not match the data:

Cluster ID	Name in exported CSV	Actual profile	Correct name
0	Frequent Buyer	Avg order 120.64, discount 0.08	High-Value Customer
1	High-Value Customer	Avg order 20.29, discount 0.77	Discount-Driven Shopper
2	Discount-Driven Shopper	Purchase freq 17.95, abandon 0.12	Frequent Buyer
3	Occasional Browser	Purchase freq 2.01, abandon 0.68	Occasional Browser (correct)

Impact: three of four labels in `ecommerce_customer_behavior_with_clusters.csv` are wrong, which would also give wrong marketing actions in the CustomerLens predictor if it reuses this mapping. **This report uses the corrected names.** Suggested fix: derive the labels from the profile table by rule (highest `avg_order_value` = High-Value; highest `discount_usage_rate` among the rest = Discount-Driven; highest `purchase_frequency` among the rest = Frequent Buyer; remaining = Occasional Browser) instead of hard-coding IDs.

8. Conclusions and Recommendations

- Data is clean and complete after median imputation; the tails in order value and discount use are genuine segment signals and were kept.
- Heavy correlation among behavioural features supports using PCA; 8 components retain 90.7% of the variance.
- t-SNE reveals four clear segments, and KMeans (k=4) recovers them with 99.5% agreement to the reference labels, giving four actionable marketing profiles.
- Be transparent in the review that the silhouette score (0.239) does not favour k=4 and that the dataset looks synthetic.
- Fix the label mapping before using the exported CSV or the CustomerLens interface.
- Possible improvements: cluster on the numeric features only (or fewer PCs) to test higher silhouette, validate on a real public dataset, and try 3D t-SNE as planned.